LAUNDRY SERVICE ORDER AND OPERATIONS MANAGEMENT SYSTEM (URBANWASH)

A PROJECT REPORT

Submitted by

PARIKSHITH P

(241401062)

in partial fulfilment of the End Semester Examination for the course

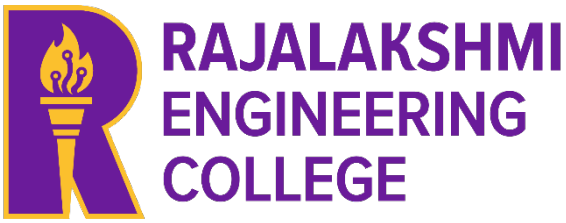
CB23F35

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE AND BUSINESS SYSTEMS



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

NOVEMBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled “Laundry Service Order and Operations Management System (UrbanWash)” is the bonafide work of Parikshith P (241401062), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, Parikshith P (241401062), hereby declare that the project report titled “Laundry Service Order and Operations Management System (UrbanWash)”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, dashboard, Deluge scripts and workflows — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 3 November 2026

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PARIKSHITH P

ABSTRACT

Small laundry businesses commonly record customers, clothing items, prices and order progress in paper registers or loosely maintained spreadsheets. Prices are quoted from memory, totals are calculated by hand, the stage of an order is known only by asking the staff, and the owner has no quick view of orders or of the revenue actually received. These practices lead to calculation errors, inconsistent customer records and delayed communication with customers.

This project presents UrbanWash, a Laundry Service Order and Operations Management System developed on the Zoho Creator low-code platform as an individual academic project. The application has three forms (Customer, Service Catalogue and Laundry Order) connected through lookup fields, five report views and a dashboard. The initial structure was generated with Zia and then configured manually. Five form workflows, written with Deluge scripting and workflow actions, retrieve the unit price of the selected service, calculate the total amount as quantity multiplied by unit price, reject delivery dates earlier than the order date, reject zero or negative quantities, and send an email notification when an order is marked Ready for Delivery. Orders move through five stages, from Received to Delivered, and payment is recorded as Paid or Pending. The dashboard gives the business owner four KPI cards and two charts.

The application was tested functionally on the live Zoho Creator interface using a demonstration dataset of five sample orders. Of sixteen defined test cases, nine were executed and passed, covering automatic pricing, total calculation, delivery-date and zero-quantity validation, order submission, the email alert, the dashboard and the reports; seven cases were not executed within the project timeline. The dashboard reported ₹330 of revenue from paid orders, correctly excluding pending orders from a total order value of ₹1,476. Three defects remain open and are documented with suggested corrections. The application is a standalone demonstration built in a Zoho Creator trial account; it uses no external integrations and is not presented as a production system. The work shows that a small business process can be digitised quickly with low-code configuration and a little Deluge scripting.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Order Management; Laundry Services*

PROJECT AT A GLANCE

Attribute	Details
Project Title	Laundry Service Order and Operations Management System (UrbanWash)
Application Domain	Service management – laundry business
Student Name & Register No.	Parikshith P – 241401062
Zoho Creator Edition	Trial account (edition not verified)
Application Name (Link Name)	urbanwash-laundry-service-order-management
Published / Live URL	https://creatorapp.zoho.in/parikshith.p.2024.csbs_ra844/urbanwash-laundry-service-order-management
Demo Credentials (per role)	demourbanwash demourbanwash@gmail.com
Number of Forms / Subforms	3
Number of Reports (by type)	5 report views: All Customers, All Service Catalogues, All Laundry Orders (List); Order Statuses, Service Categories (Kanban)
Number of Pages / Dashboards	1 dashboard (4 KPI cards, 2 charts, embedded report, quick links)
Number of Workflows	Form: 5 Schedule: 0 Approval: 0 Blueprint: 0 Batch: 0
Deluge Functions / Custom Actions	4 Deluge-scripted form workflows
Integrations	None
Roles & Profiles Defined	One role configured (CEO); permission entries: Business Owner, Front Desk Staff, Read, Write
Environments Used	Development only (trial account); no Stage or Production environment is claimed
Date of Production Deployment	Demonstration deployment application created on 31 October 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
CEO	Chief Executive Officer (the configured role name)
CO	Course Outcome
CRUD	Create, Read, Update, Delete
Deluge	Data Enriched Language for the Universal Grid Environment
ER	Entity–Relationship
FR	Functional Requirement
INR	Indian Rupee
KPI	Key Performance Indicator
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
TC	Test Case
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho's AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Laundry services are a common small-business activity in which customers hand over clothes for washing, dry cleaning or ironing and collect them after a promised delivery date. A typical order passes through several stages: the clothes are received, washed, ironed, made ready for delivery and finally delivered, and the customer pays either at drop-off or at collection. The front desk staff register customers and orders, while the owner is interested in how many orders are active and how much revenue has actually been received.

UrbanWash is an academic application modelled on such a small laundry business. It was built on Zoho Creator to demonstrate how forms, reports, a dashboard, Deluge scripts and workflows can digitise this process. The application is not deployed at a real business, and all records shown in this report are sample data created for demonstration and testing.

1.2 Problem Statement

Small laundry businesses often manage customers, services, prices and orders on paper or in spreadsheets. As a result, staff must look up prices and calculate totals manually, customer details are repeated in every order, the progress of an order is not visible to the owner, and revenue received cannot be separated easily from revenue still pending. A simple, structured and automated order management system is needed. The main pain points are:

- Customer and service details are entered repeatedly, causing spelling differences and duplicate records.
- Prices and totals are calculated manually, which can lead to wrong bills.
- There is no live view of order status, so the owner and customers must ask staff for progress.
- Paid and pending orders are not separated, so received revenue is hard to determine.
- Customers are not informed automatically when their laundry is ready.

1.3 Objectives

- To design a three-form data model (Customer, Service Catalogue and Laundry Order) linked through lookup fields.
- To automate unit-price retrieval and total-amount calculation using Deluge scripting and form workflows.
- To validate the expected delivery date and the clothing quantity before an order is saved.
- To notify staff by email when an order is marked Ready for Delivery.
- To provide reports and a dashboard that show paid revenue, order volume, order status distribution and service categories.

1.4 Scope of the Project

In scope: a three-form Zoho Creator application for a single laundry business; customer registration; a service catalogue with three categories (Washing, Dry Cleaning and Ironing); order creation with automatic unit price and total amount; delivery-date and quantity validation; tracking of order status through five stages and of payment status (Paid or Pending); an email alert for Ready for Delivery; five report views; and a dashboard with four KPI cards and two charts. The intended users are the Business Owner and the Front Desk Staff, and the application is used through the web interface.

Out of scope: integration with external systems or other Zoho services, online payment, SMS notifications, a customer portal, inventory and staff management, mobile-specific layouts, and any pro-code extension. A separate payment form is not used; payment status is stored in the Laundry Order form. Role-based permission profiles were not part of the tested scope of this version.

1.5 Justification for a Low-Code Approach

Low-code platforms allow business applications to be assembled mostly through visual configuration, with little hand-written code to maintain [1]. Zoho Creator provides built-in data storage, forms, reports, dashboards, workflow automation and cloud hosting, so no separate database or server had to be designed [2]. Zoho Creator was selected because it is the platform prescribed for course CB23F35 and provides all of these capabilities in one hosted environment. Practitioner studies report that low-code

platforms are easy to learn, speed up development and are favoured where processes and workflows need to be automated [3].

These characteristics fit UrbanWash well. The application is data-entry centred, has a small number of entities, and needs a few business rules (pricing, totals, validation and notification). It was therefore practical for one student to build, test and demonstrate the complete application on Zoho Creator, using Deluge only for the few rules that configuration alone could not express.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

No real business was surveyed for this project; the existing system described here is the manual process that is typical of small laundries and that UrbanWash is designed to replace. The customer hands over clothes at the counter. The staff member writes the customer name, items and charge in a paper register or notebook, estimates the price from memory or a printed rate card, and gives the customer a delivery date. The status of the order is known only by asking the staff, the customer calls or visits to ask about progress, and the owner tallies payments by hand at the end of the day or month. Figure 2.1 shows this as-is flow.

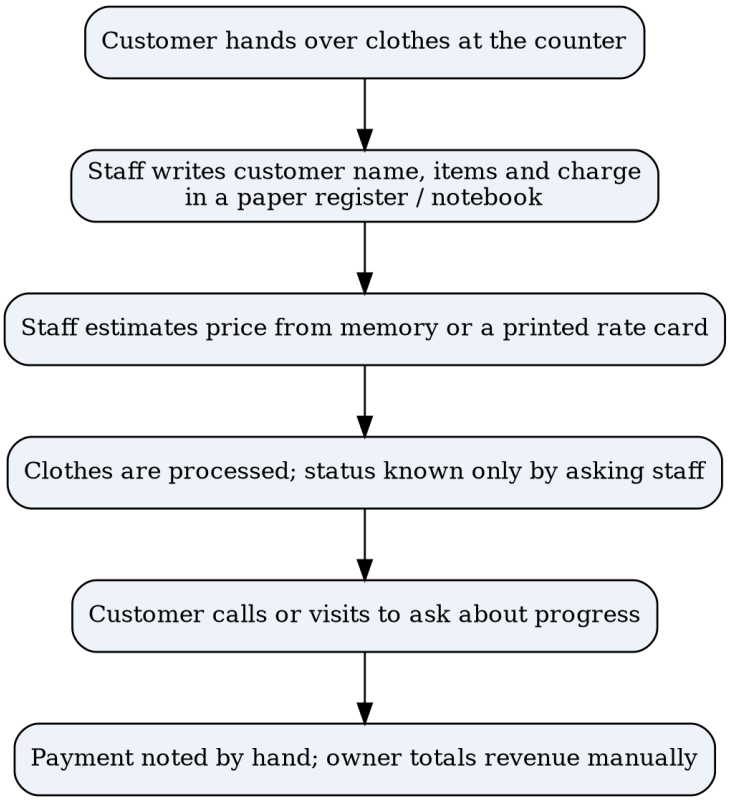


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Customer details are rewritten for every order, so records are duplicated and inconsistent.

- Prices are quoted from memory or a rate card, and totals are calculated by hand, which causes billing errors.
- The status of an order cannot be seen without asking the staff.
- Revenue received cannot be separated quickly from revenue pending, and reports must be tallied manually.
- Customers are not notified automatically when an order is ready.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development effort	Design a database, back-end, front-end and hosting before the first screen works	Forms, reports and a dashboard are configured visually; the first version of UrbanWash was generated with Zia and then refined
Skills required	Programming, database design and web development	Data modelling and basic Deluge scripting
Hosting & infrastructure	Server or cloud hosting must be arranged and maintained	Hosted by Zoho Creator; no server set-up
Mobile application	Separate mobile application to be developed	Not developed for UrbanWash; Zoho Creator provides a mobile app that was not used here
Security & access control	Authentication and permissions coded manually	Zoho account login is built in; a CEO role and permission entries (Business Owner, Front Desk Staff, Read, Write) are defined; their assignment and enforcement were not verified
Maintenance & change	Code changes, testing and redeployment	Fields, reports and workflows can be edited in the builder
Customisation limits	Unlimited, at the cost of effort	Limited to what the platform and Deluge support
Cost	Developer time, hosting and licences	Platform subscription (or education/trial access) and the developer’s time

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	R. Sanchis et al., Applied Sciences, 2020 [1]	Low-Code as Enabler of Digital Transformation in Manufacturing Industry	Low-code platforms facilitate and automate application development and leave little code to maintain; platforms are benchmarked for missing features	Supports the maintainability argument and the discussion of platform limits
2	Zoho Corporation [2]	Zoho Creator Help Documentation	Describes forms, reports, pages, workflows and the platform's features	Reference for configuring UrbanWash's forms, workflows and dashboard
3	Y. Luo et al., ESEM 2021 [3]	Characteristics and Challenges of Low-Code Development: The Practitioners' Perspective	Low-code platforms are easy to learn, speed up development and suit automated processes and workflows; practitioners hold mixed views of its limits	Supports the choice of low-code for a workflow-centred order application
4	R. Waszkowski, IFAC-PapersOnLine, 2019 [4]	Low-code platform for automating business processes in manufacturing	Applies a low-code BPM platform to automate business processes	Shows a business-process use of low-code similar to UrbanWash's order workflow
5	Zoho Corporation [5]	Deluge Script Reference Guide	Describes the Deluge scripting language used in Zoho Creator form events and workflows	Reference for the four Deluge scripts used in UrbanWash

2.5 Emerging Trends in Low-Code Development

Several trends in low-code development influenced UrbanWash. First, AI-assisted app building: the initial forms, reports and dashboard were generated by describing the business in a prompt to Zia and were then reviewed and corrected manually. Second, citizen development: low-code platforms let business users with limited programming experience assemble applications [1], [3], which is why UrbanWash relies mostly on

configuration and uses Deluge only for pricing, totals and validation. Third, hyper-automation: routine steps such as price lookup, total calculation and the ready-for-delivery email are automated through workflows. Finally, composable applications: the three-form structure could be extended later with a payment form or integrations without redesigning the existing forms.

2.6 Proposed System

UrbanWash is a Zoho Creator application that replaces the manual register with three linked forms. Customers are stored once and selected through a lookup, which removes repeated entry (Limitation 1). The Service Catalogue holds the price of each service, and the Auto Price and Calculate Total workflows fill the unit price and the total, which removes manual pricing (Limitation 2). The Order Status field and the Order Statuses report make progress visible (Limitation 3). The dashboard computes revenue from orders marked Paid, separately from pending orders (Limitation 4), and the Ready for Delivery Alert workflow sends an email when an order is ready (Limitation 5).

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Business Owner	Owns the laundry business and monitors its performance	View dashboard KPIs and charts; review orders and revenue; receive ready-for-delivery emails
Front Desk Staff	Handles customers at the counter	Register customers; maintain services; create orders; update order and payment status
Customer	External party; does not log in to the system	Hands over and collects clothes; no customer portal in this version

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow staff to register a customer with name, phone number, email, address and registration date, and assign a unique Customer ID.	Staff	High
FR-02	The system shall maintain a service catalogue with service name, category (Washing, Dry Cleaning, Ironing), price per unit and description.	Staff	High
FR-03	The system shall allow staff to create a laundry order by selecting a customer and a service through lookup fields.	Staff	High
FR-04	The system shall populate the Unit Price automatically from the selected service.	Staff	High
FR-05	The system shall calculate Total Amount as Clothing Quantity × Unit Price.	Staff	High
FR-06	The system shall reject an order whose Expected Delivery Date is earlier than the Order Date.	Staff	High
FR-07	The system shall reject an order whose Clothing Quantity is zero or negative.	Staff	High
FR-08	The system shall track Order Status (Received, Washing, Ironing, Ready for Delivery, Delivered) and Payment Status (Paid, Pending).	Staff	High
FR-09	The system shall send an email notification when an order is marked Ready for Delivery.	Owner, Staff	Medium

FR ID	Requirement	Role(s)	Priority
FR-10	The system shall display a dashboard with paid revenue, total orders, orders ready for delivery, orders in progress and charts of orders by status and services by category.	Owner	Medium
FR-11	The system shall provide reports of customers, services, orders and order statuses.	Owner, Staff	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and the dashboard for a small laundry business (a few hundred records) shall load without noticeable delay.
NFR-02	Security	Only authenticated Zoho users shall be able to open the application.
NFR-03	Usability	The application shall be usable through a standard web browser; entering an order shall require no manual arithmetic.
NFR-04	Availability	The application shall be hosted on the Zoho Creator cloud so that no server needs to be maintained by the business.
NFR-05	Maintainability	Business rules shall be implemented as named workflows so that they can be changed without rebuilding the application.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator trial account (edition not verified)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Not used (the application has no API integrations)
Other Zoho services	None (the email alert uses Zoho Creator’s built-in Send Mail action)
Mobile	Not used (web interface only)
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

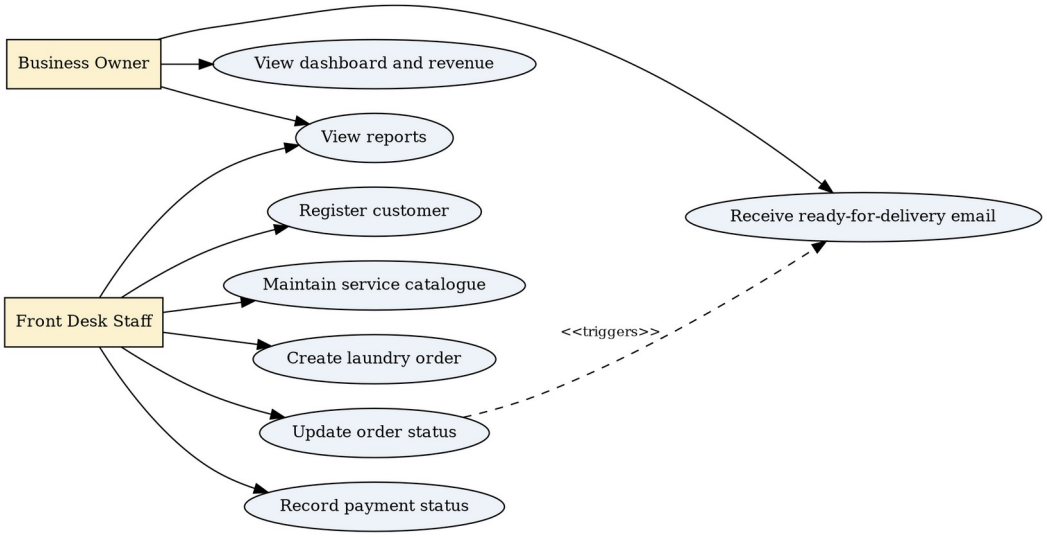


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Create Laundry Order

Use Case ID	UC-01
Use Case Name	Create Laundry Order
Actor(s)	Front Desk Staff
Pre-condition	The customer and the service already exist in the Customer and Service Catalogue forms.
Main Flow	1. Staff opens the Laundry Order form. 2. Staff selects the Customer and the Service. 3. The Auto Price workflow fills the Unit Price. 4. Staff enters Order Date, Expected Delivery Date and Clothing Quantity. 5. The Calculate Total workflow fills the Total Amount. 6. Staff selects Order Status and Payment Status and submits the form.
Alternate Flow	If the Expected Delivery Date is earlier than the Order Date, or the Clothing Quantity is zero or negative, an alert is shown and submission is cancelled; staff correct the value and resubmit.
Post-condition	The order is saved and appears in All Laundry Orders and in the dashboard figures.
Related FR	FR-03, FR-04, FR-05, FR-06, FR-07

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

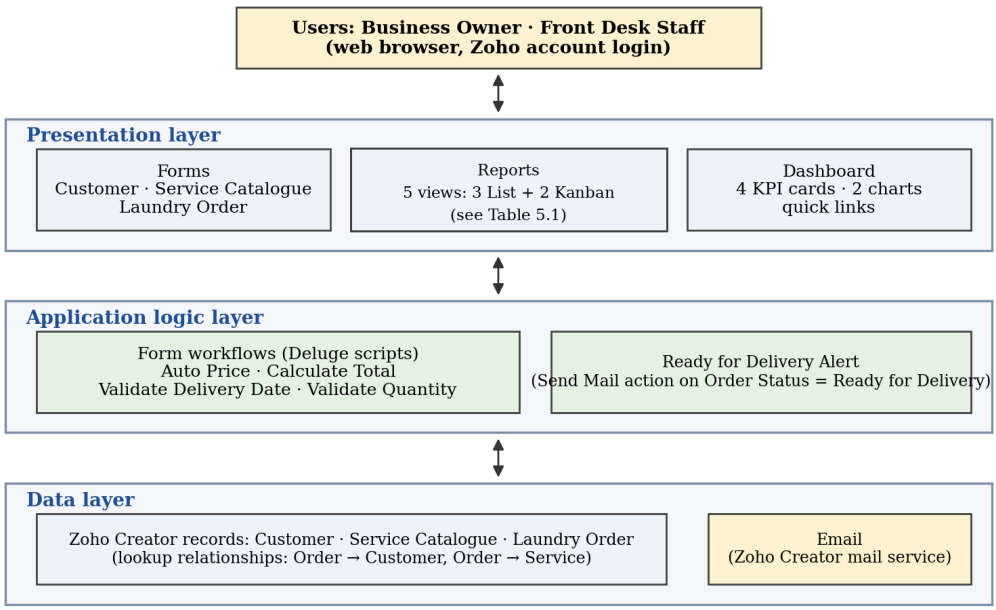


Figure 4.1: System Architecture of the Application

UrbanWash follows a simple three-layer logical structure, shown in Figure 4.1. Users (the Business Owner and Front Desk Staff) sign in with a Zoho account and use the application through a web browser. The presentation layer consists of the three forms, the five report views and the dashboard. The application logic layer consists of the form workflows: Auto Price, Calculate Total, Validate Delivery Date and Validate Quantity, which use Deluge scripts, and the Ready for Delivery Alert, which uses a Send Mail action. The data layer is the set of records stored by Zoho Creator for the Customer, Service Catalogue and Laundry Order forms. This is a logical description; no separate servers or database infrastructure were configured, and there are no external integrations.

4.2 Data Model

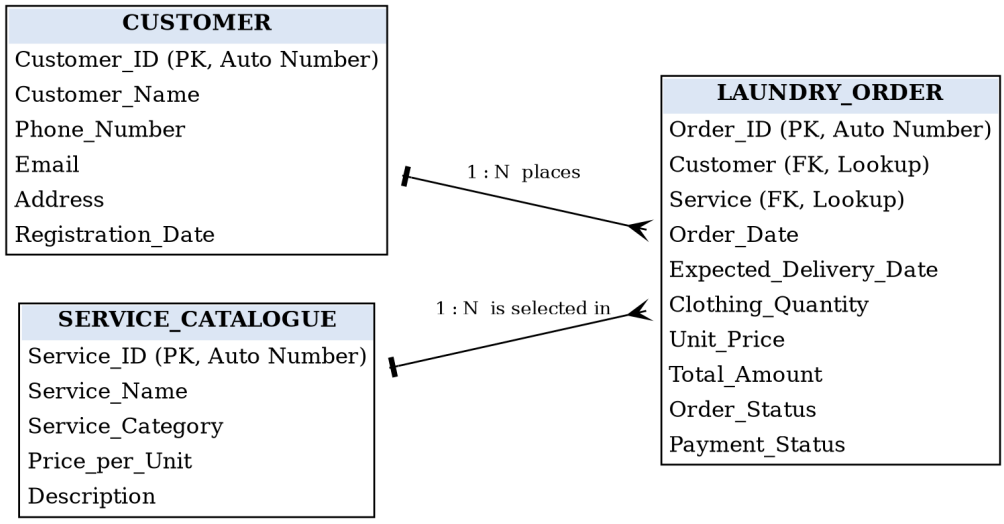


Figure 4.2: Entity–Relationship Diagram

The data model has three entities. Customer stores registration and contact details. Service Catalogue stores each laundry service with its category and price per unit. Laundry Order is the transaction form: it refers to one customer and one service through lookup fields, and stores the dates, quantity, unit price, total amount, order status and payment status. One customer can place many orders and one service can appear in many orders, so both relationships are one-to-many. The lookups allow a single selection only, and no subforms are used. Lookup fields were used instead of typed names so that each customer and service is stored once and selected from existing records; this avoids spelling differences and keeps each price in the Service Catalogue, from which the Auto Price workflow retrieves it.

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Customer	Customer ID, Customer Name, Email	Laundry Order	One-to-Many	Lookup field Customer in Laundry Order
Service Catalogue	Service ID, Service Name, Price per Unit	Laundry Order	One-to-Many	Lookup field Service in Laundry Order
Laundry Order	Order ID, Order Date, Order Status	Customer; Service Catalogue	Many-to-One (to each)	Two single-selection lookup fields

4.3 Form Design

Table 4.2: Field Specification – UrbanWash Forms

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Customer form					
Customer ID	Customer_ID	Auto Number	Auto	Unique customer identifier	Primary key
Customer Name	Customer_Name	Name	Yes	—	—
Phone Number	Phone_Number	Phone	Yes	—	—
Email	Email	Email	No	—	—
Address	Address	Address	No	—	—
Registration Date	Registration_Date	Date	Yes	—	—
Service Catalogue form (link name: Service_Catalogue)					
Service ID	Service_ID	Auto Number	Auto	Unique service identifier	Primary key
Service Name	Service_Name	Single Line	Yes	—	—
Service Category	Service_Category	Drop Down	Yes	Washing / Dry Cleaning / Ironing	—
Price per Unit	Price_per_Unit	Currency	Yes	Source of the unit price	INR
Description	Description	Multi Line	No	—	—
Laundry Order form					
Order ID	Order_ID	Auto Number	Auto	Unique order identifier	Primary key
Customer	Customer	Lookup	Yes	Refers to Customer	Single selection; displays Customer ID
Service	Service	Lookup	Yes	Refers to Service Catalogue	Single selection; displays Service Name; new entries not allowed
Order Date	Order_Date	Date	Yes	Compared with delivery date	—

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Expected Delivery Date	Expected_Delivery_Date	Date	Yes	Must not be earlier than Order Date	Validate Delivery Date
Clothing Quantity	Clothing_Quantity	Number	Yes	Must be greater than zero	Validate Quantity
Unit Price	Unit_Price	Currency	Yes	Filled automatically from the service	INR, 2 decimal places
Total Amount	Total_Amount	Currency	Yes	Quantity × Unit Price (filled by workflow)	INR
Order Status	Order_Status	Drop Down	Yes	Received / Washing / Ironing / Ready for Delivery / Delivered	—
Payment Status	Payment_Status	Drop Down	Yes	Paid / Pending	—

4.4 Process and Workflow Design

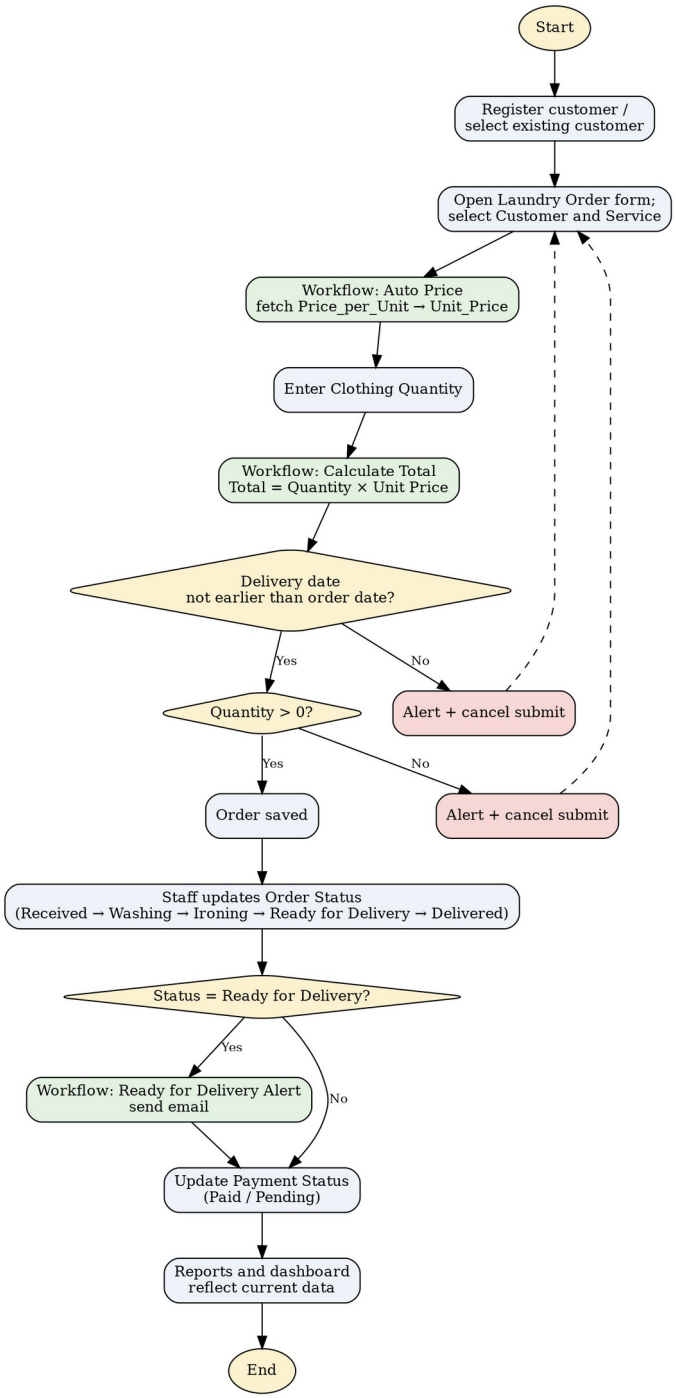


Figure 4.3: Workflow / Process Flow for Laundry Order Processing

Figure 4.3 shows how an order flows through the application. The green steps are the workflows. Auto Price runs when a service is selected and Calculate Total runs when the quantity is entered. The two validation workflows run when the form is submitted and cancel the submission with an alert if the delivery date or quantity is invalid. After the order is saved, staff update the Order Status; when it becomes Ready for Delivery,

the Ready for Delivery Alert workflow sends an email. Payment Status is then updated and the reports and dashboard reflect the current data.

4.5 User Interface and Navigation Design

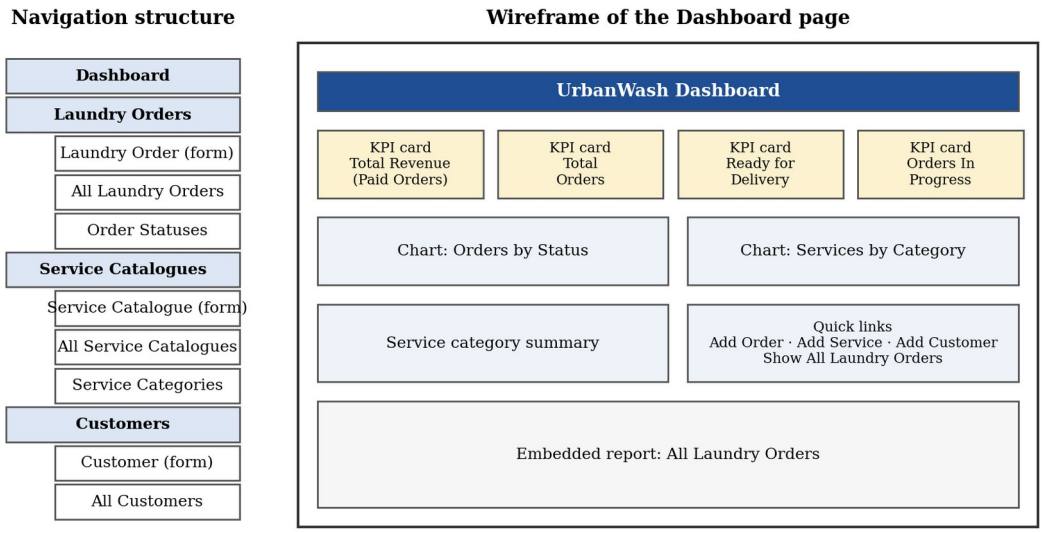


Figure 4.4: Navigation Structure and Wireframe of Dashboard

The left side of Figure 4.4 shows the navigation sidebar of the application: Dashboard, Laundry Orders (form, All Laundry Orders and Order Statuses), Service Catalogues (form, All Service Catalogues and Service Categories) and Customers (form and All Customers). The right side is a wireframe of the dashboard: four KPI cards at the top, the two charts and the service category summary in the middle, quick links to add records, and the embedded All Laundry Orders report at the bottom.

4.6 Access Control Design

Table 4.3: Role–Permission Matrix

Component	CEO (role)	Business Owner (permission entry)	Front Desk Staff (permission entry)	Portal User
Form: Customer	Not verified	V	AEV	—
Form: Laundry Order	Not verified	V	AEV	—
Report: All Laundry Orders	Not verified	V	V	—
Page: Dashboard	Not verified	V	Not verified	—

The matrix is derived from the permission descriptions shown in the application (Business Owner: views all data, does not need to create or update records; Front Desk

Staff: full access to create and update customers, service catalogue entries and laundry orders). A = Add, E = Edit, V = View; Not verified = not confirmed in the application. Only one role, CEO, is configured; the two entries are permission definitions, and their assignment to users and enforcement were not verified. No customer portal exists, so Portal User has no access.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application, named UrbanWash Laundry Service Order Management, was created in Zoho Creator using the Create using Zia option. A detailed prompt described the laundry business, the customers, services and laundry orders, the order and payment statuses, the reports and dashboard, the Business Owner and Front Desk Staff personas and a professional blue-and-white interface. Zia generated the initial forms, reports, dashboard and sample data. The generated application was then inspected and changed manually: the currency was changed to Indian Rupees, the lookup fields were configured, the workflows were created and the dashboard was reconfigured. The sidebar sections are Dashboard, Laundry Orders, Service Catalogues and Customers.

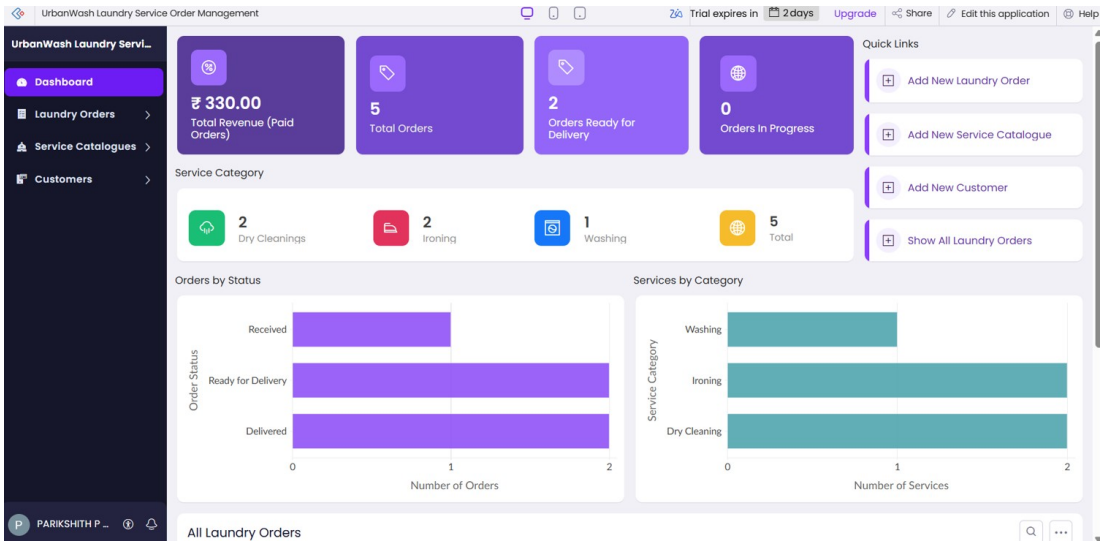


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

The application has three primary forms. The Customer form holds the customer's name, phone number, email, address and registration date, with an auto-numbered Customer ID; Customer Name, Phone Number and Registration Date are mandatory, while Email and Address are optional. The Service Catalogue form holds the service name, a Service Category drop-down (Washing, Dry Cleaning or Ironing), the price per unit and a description, with an auto-numbered Service ID; all fields except Description are mandatory. The Laundry Order form is the main transaction form and all its fields are mandatory. Its Customer and Service fields are single-selection lookups. The

Service lookup reads from the Service Catalogue, displays the Service Name as a drop-down, is mandatory, is sorted alphabetically and does not allow new entries. The Customer lookup displays the Customer ID, so customers appear as numbers in the form and reports (Section 6.4). Order Date and Expected Delivery Date are Date fields, Clothing Quantity is a Number field, and Price per Unit, Unit Price and Total Amount are currency fields in Indian Rupees (INR). Order Status and Payment Status are drop-downs. Unit Price and Total Amount are filled by workflows, and the delivery date and quantity are checked by validation workflows (Section 5.6).

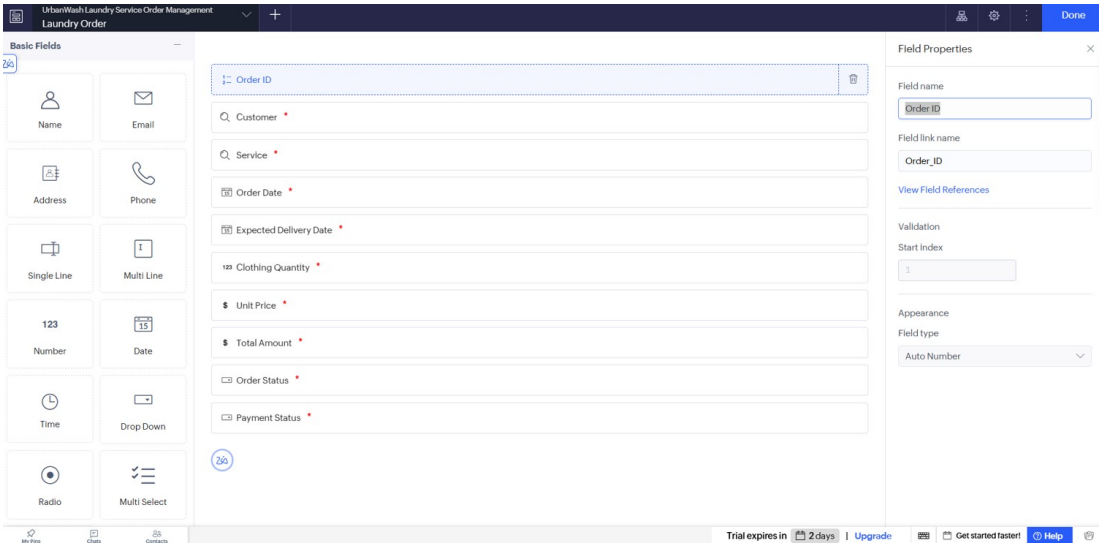


Figure 5.2: Laundry Order Form in the Form Builder

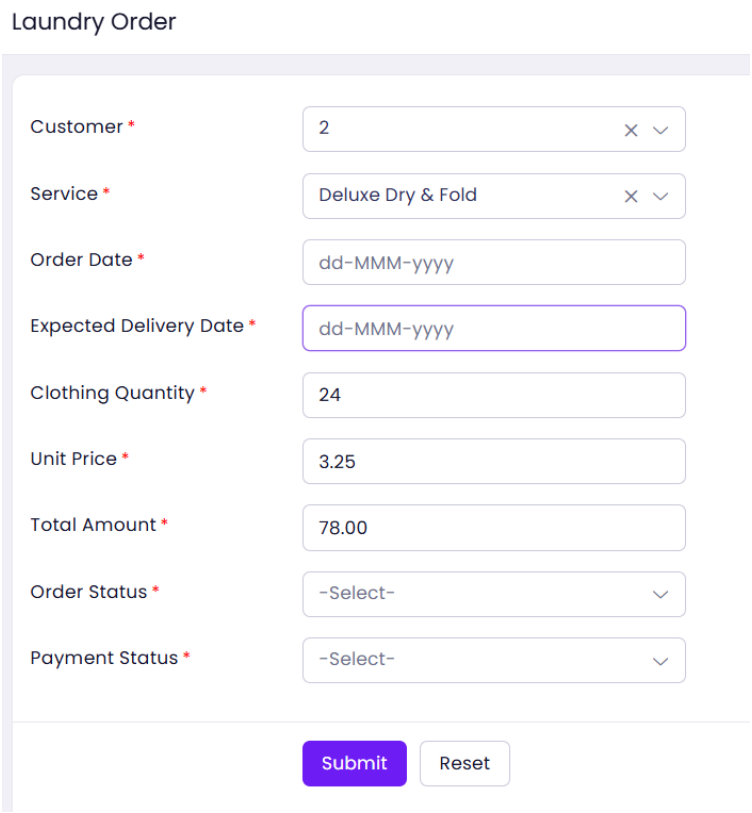
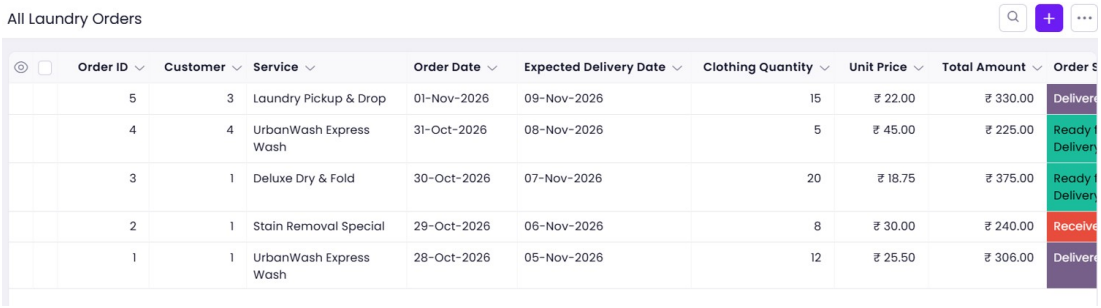


Figure 5.3: Laundry Order Form in Live Mode

5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Customers	List	Customer	All records	—	Not verified
All Service Catalogues	List	Service Catalogue	All records	—	Not verified
All Laundry Orders	List	Laundry Order	All records; lookups, dates, quantity, prices, total and statuses	—	Not verified
Order Statuses	Kanban	Laundry Order	Grouped by Order Status (columns)	—	Not verified
Service Categories	Kanban	Service Catalogue	Grouped by Service Category (columns)	—	Not verified



Order ID	Customer	Service	Order Date	Expected Delivery Date	Clothing Quantity	Unit Price	Total Amount	Order S
5	3	Laundry Pickup & Drop	01-Nov-2026	09-Nov-2026	15	₹ 22.00	₹ 330.00	Deliver
4	4	UrbanWash Express Wash	31-Oct-2026	08-Nov-2026	5	₹ 45.00	₹ 225.00	Ready f Deliver
3	1	Deluxe Dry & Fold	30-Oct-2026	07-Nov-2026	20	₹ 18.75	₹ 375.00	Ready f Deliver
2	1	Stain Removal Special	29-Oct-2026	06-Nov-2026	8	₹ 30.00	₹ 240.00	Receive
1	1	UrbanWash Express Wash	28-Oct-2026	05-Nov-2026	12	₹ 25.50	₹ 306.00	Deliver

Figure 5.4: All Laundry Orders Report

5.4 Pages and Dashboards

UrbanWash has one dashboard, configured manually after Zia generated the first version. It has four KPI cards: Total Revenue (Paid Orders), which sums Total Amount of orders whose Payment Status is Paid; Total Orders, which counts all Laundry Order records; Orders Ready for Delivery, which counts orders whose Order Status is Ready for Delivery; and Orders In Progress, which counts orders whose Order Status is Washing only and therefore excludes Ironing (known limitation BUG-04, Table 6.2). Two charts are shown: Orders by Status (count of Order ID against Order Status) and Services by Category (count of Service ID against Service Category). The page also has a service category summary, an embedded All Laundry Orders report and quick links to add a Laundry Order, Service Catalogue entry or Customer and to show all laundry orders. With the sample data, the live dashboard showed ₹330 revenue, 5 orders, 2 ready for delivery and 0 in progress.

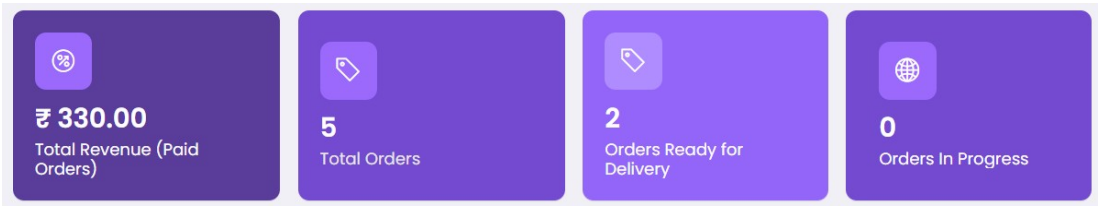


Figure 5.5: UrbanWash Dashboard

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Laundry Order → On User Input of Service	Fetch the service price, fill Unit Price and refresh Total Amount	fetch by ID, if, null check, field assignment

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-02	Laundry Order → On User Input of Clothing Quantity	Calculate Total Amount = quantity × unit price	if, null check, arithmetic
DS-03	Laundry Order → On Validate	Reject an expected delivery date earlier than the order date	if, comparison, alert, cancel submit
DS-04	Laundry Order → On Validate	Reject zero or negative clothing quantity	if, comparison, alert, cancel submit

Code 5.1: Auto Price script (DS-01) – Laundry Order, Created or Edited → User input of Service

```
if(input.Service != null)
{
    serviceRecord = Service_Catalogue[ID == input.Service];
    if(serviceRecord != null)
    {
        input.Unit_Price = serviceRecord.Price_per_Unit;
        if(input.Clothing_Quantity != null)
        {
            input.Total_Amount = input.Clothing_Quantity *
input.Unit_Price;
        }
    }
}
```

Code 5.2: Calculate Total script (DS-02) – User input of Clothing Quantity

```
if(input.Clothing_Quantity != null && input.Unit_Price != null)
{
    input.Total_Amount = input.Clothing_Quantity * input.Unit_Price;
}
```

Code 5.3: Validate Delivery Date (DS-03) and Validate Quantity (DS-04) – Validations on form submission

```
if(input.Expected_Delivery_Date < input.Order_Date)
{
    alert "Expected Delivery Date cannot be earlier than Order Date.";
    cancel submit;
}

if(input.Clothing_Quantity <= 0)
{
    alert "Clothing Quantity must be greater than zero.";
    cancel submit;
}
```

DS-01 (Auto Price) runs when the user changes the Service field. It fetches the selected record from Service_Catalogue, copies Price_per_Unit into Unit_Price and, if a quantity has already been entered, recalculates Total_Amount. DS-02 (Calculate Total) runs when the user changes Clothing Quantity and multiplies the quantity by the unit

price. DS-03 and DS-04 run on form submission; if the delivery date is earlier than the order date, or the quantity is zero or negative, they show an alert and cancel the submission. DS-01 to DS-04 were observed working on the live form for the cases recorded in Table 6.1; the negative-quantity case (TC-11) was not executed. Deluge is described in [5]. The scripts shown were checked against screenshots of the saved workflows.

5.6 Workflows and Automation

Workflow events decide when a script runs. A User Input event runs while a field is being filled (Auto Price on Service; Calculate Total on Clothing Quantity). An On Validate event runs when the form is submitted, before the record is saved, and can cancel the submission with an alert (Validate Delivery Date and Validate Quantity). A Successful Form Submission event runs after the record has been saved; the Ready for Delivery Alert uses it with the condition Order Status equals Ready for Delivery, so mail is sent only when an edited order reaches that status.

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Auto Price	Form Workflow	On User Input of Service	Service is not empty	Fetch price, set Unit Price, refresh Total Amount	DS-01
Calculate Total	Form Workflow	On User Input of Clothing Quantity	Quantity and Unit Price are not empty	Set Total Amount	DS-02
Validate Delivery Date	Form Workflow	On Validate (created or edited)	Delivery date earlier than order date	Alert and cancel submit	DS-03
Validate Quantity	Form Workflow	On Validate (created or edited)	Quantity is zero or negative	Alert and cancel submit	DS-04
Ready for Delivery Alert	Form Workflow	On successful submission of an edited record	Order Status = Ready for Delivery	Send Mail: “UrbanWash – Order Ready for Delivery”	—

The Ready for Delivery Alert is a form workflow on the Laundry Order form that runs on successful submission of an edited record when Order Status equals Ready for

Delivery. Its Send Mail action uses zoho.adminuserid as the sender and sends to zoho.loginuserid and one additional configured college email address, with the subject “UrbanWash – Order Ready for Delivery”. The message body gives the Order ID, Customer and Expected Delivery Date, inserted with the Insert Fields option. The email was received during testing (Figure 8.2); the Customer value appears as the lookup ID rather than the customer name (BUG-05, Table 6.2).

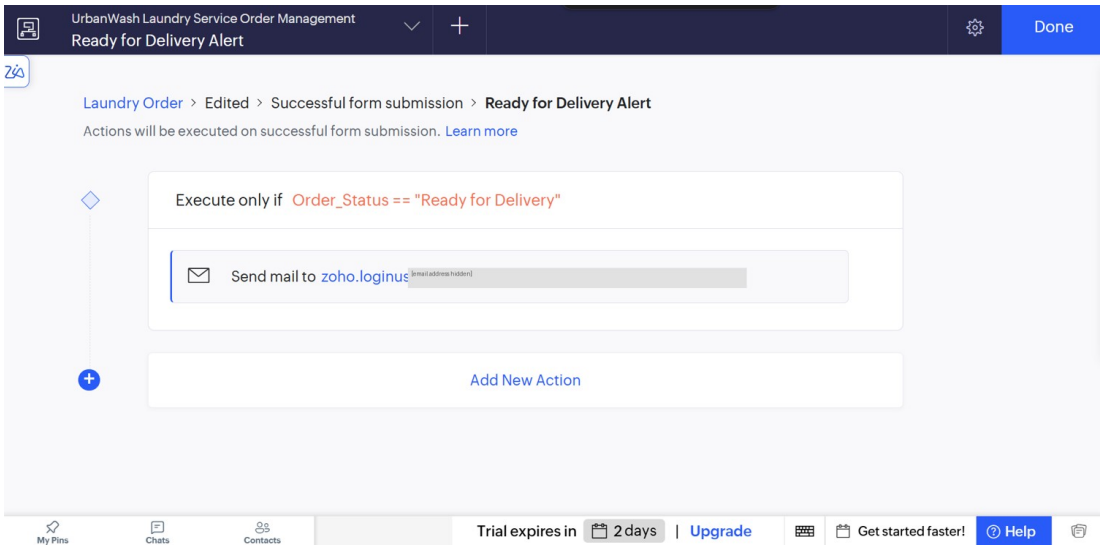


Figure 5.6: Ready for Delivery Alert Workflow Configuration

5.7 Integrations

No integration with other Zoho services or external systems was implemented, and no external API was used. UrbanWash is a self-contained application: all data is entered and stored in its three forms, and the only outgoing communication is the email alert sent by Zoho Creator’s built-in Send Mail action. Invokeurl calls, connections, webhooks and the Creator REST API were not required for this project.

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
None	—	—	—	UrbanWash is a standalone application

5.8 User Management and Role-Based Access

In the application's Users section, the Roles tab holds one configured role, CEO. The Permissions tab lists four permission entries: Business Owner, Front Desk Staff, Read and Write. Business Owner is described as monitoring business performance and viewing dashboards and reports without needing to create or update records, and Front Desk Staff as registering customers, creating orders, updating order status and recording payment status with full access to the customer, service catalogue and order forms. Read and Write grant read or write permission for all components. The report does not claim that these permissions were assigned to individual users or that their enforcement was tested; Business Owner and Front Desk Staff are permission definitions, not separate roles.

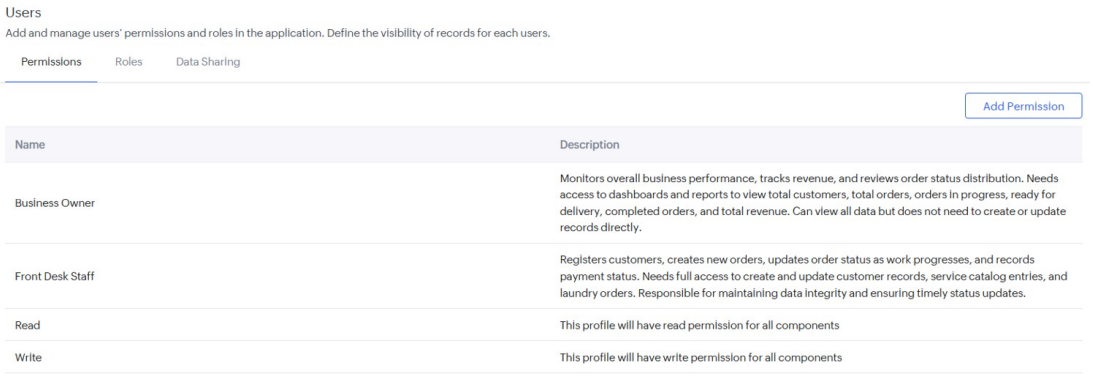


Figure 5.7: Roles and Permission Profiles Configuration

5.9 AI and Analytics Features

Zia was used once, at the start, through the Create using Zia option, to generate the first version of the forms, reports, dashboard and sample data from a written description of the business. This is AI-assisted application building; the delivered application is not an AI application and contains no prediction, OCR or other AI field. The generated output was not accepted as it stood: the currency, lookups, workflows and dashboard were configured or corrected manually and then checked on the live application (for example, chart titles that did not match their data were corrected). The sample records were also generated by Zia and are test data, not real business data.

For analytics, the dashboard provides four KPI cards and two charts built with Zoho Creator's dashboard builder. No pivot report or Zoho Analytics integration was used.



Figure 5.8: Dashboard Charts (Analytics)

5.10 Pro-Code Extension

No pro-code extension (Zoho Catalyst, custom functions, JavaScript widgets or the Creator REST API) was used. All of UrbanWash's rules—price lookup, total calculation, validation and notification—could be implemented with form workflows and short Deluge scripts. A pro-code extension would become worthwhile if the business needed heavy processing, a custom user interface or a connection to an external payment or messaging service.

5.11 Mobile and Portal Access

UrbanWash was developed and demonstrated through the web interface. Zoho Creator applications can also be opened in the Zoho Creator mobile app, but no mobile-specific layout or offline setting was configured and the application was not tested on mobile in this project. No customer portal was created; customers do not log in to the system.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was manual and functional, carried out on the live Zoho Creator application with sample data. At unit level, the Laundry Order form and its workflows were exercised by entering values on the live form. At integration level, the chain of lookup, automatic price, total calculation, validation, saving and reporting was checked, together with the email notification and the dashboard figures. Negative tests were used for the two validations. Role-based access testing was not carried out because permission assignments were not verified for this version. Table 6.1 lists sixteen test cases: nine were executed and passed, and seven were not executed within the project timeline and are marked Not Executed. Preliminary feedback from two individuals is described in Section 6.5; it is not a formal user acceptance test.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Register a customer with valid details	Open Customer form; enter name, phone, email, address, date; submit	Record saved and listed in All Customers	Not executed within the project timeline	Not Executed
TC-02	FR-02	Add a service to the catalogue	Open Service Catalogue form; enter name, category, price; submit	Record saved and listed in All Service Catalogues	Not executed within the project timeline	Not Executed
TC-03	FR-03	Create an order using the lookups	Choose Customer and Service; complete the form; submit	Order saved with a success message	“Laundry Order Added Successfully” (screenshot 225202)	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-04	FR-04	Automatic unit price	Select Service “Deluxe Dry & Fold”	Unit Price filled from the catalogue	Unit Price 3.25 filled (224516)	Pass
TC-05	FR-05	Total calculation	Quantity 24 with Unit Price 3.25	Total Amount = 78.00	Total Amount 78.00 (224523)	Pass
TC-06	FR-05	Recalculate total when quantity changes	Change quantity of an entered order	Total Amount updates	Not executed within the project timeline	Not Executed
TC-07	FR-04, FR-05	Change service after entering quantity	Change the Service field	Unit Price and Total Amount update	Not executed within the project timeline	Not Executed
TC-08	FR-06	Delivery date earlier than order date	Expected Delivery Date before Order Date; submit	Alert shown; submission cancelled	Alert “Expected Delivery Date cannot be earlier than Order Date.” (225052)	Pass
TC-09	FR-06	Delivery date equal to order date	Expected Delivery Date = Order Date; submit	Order accepted	Not executed within the project timeline	Not Executed
TC-10	FR-07	Zero quantity	Enter quantity 0; submit	Alert shown; submission cancelled	Alert “Clothing Quantity must be greater than zero.” (225108)	Pass
TC-11	FR-07	Negative quantity	Enter quantity -2; submit	Alert shown; submission cancelled	Not executed within the project timeline	Not Executed
TC-12	FR-08	Update order and payment status	Edit an order; change Order Status and Payment Status	Changes saved and shown in reports	Not executed within the project timeline	Not Executed

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-13	FR-09	Ready for Delivery email	Edit an order and set Order Status to Ready for Delivery	Email “UrbanWash – Order Ready for Delivery” received	Email received with Order ID 3, Customer 1 (ID, not name), Expected Delivery Date 07-Nov-2026 (224757, 225228); first message was in Spam	Pass
TC-14	FR-10	Dashboard KPIs and charts	Open the live dashboard	Actual values shown instead of placeholders	₹330.00, 5, 2 and 0; both charts shown (224328, 224342, 224351)	Pass
TC-15	FR-10	Revenue counts only paid orders	Compare dashboard revenue with the order report	Only Payment Status = Paid is summed	Revenue ₹330 equals Order 5, the only Paid order (224342, 224554)	Pass
TC-16	FR-11	Reports show the expected records	Open the five report views	Records displayed as specified	All five views displayed records (224416, 224433, 224438, 224446, 224450)	Pass

Summary of results: 16 test cases were defined. Nine were executed and passed (TC-03, TC-04, TC-05, TC-08, TC-10, TC-13, TC-14, TC-15 and TC-16), none failed, and seven were not executed (TC-01, TC-02, TC-06, TC-07, TC-09, TC-11 and TC-12). No outcome has been recorded for the cases that were not executed. Numbers in brackets in the Actual Result column (for example 225202) are the time stamps of the screenshot files kept as evidence (Screenshot 2026-11-01 HHMMSS); they are not figure numbers.

6.3 Debugging Deluge Scripts

Scripts were debugged by running them on the live form and observing the result: the alert messages of the two validation workflows show when a rule fires, and the values filled into Unit Price and Total Amount show whether the pricing workflows ran. The dashboard was checked in the live application, because the dashboard designer shows only placeholders such as Sum() and Ct() instead of real values. One real problem encountered was that the first ready-for-delivery email did not appear in the inbox; it had been delivered to Gmail Spam, as shown by its sender, notifications@trial.zohocreatormail.com. After the message was found, an order was edited from the report and the email arrived with the field values filled in.

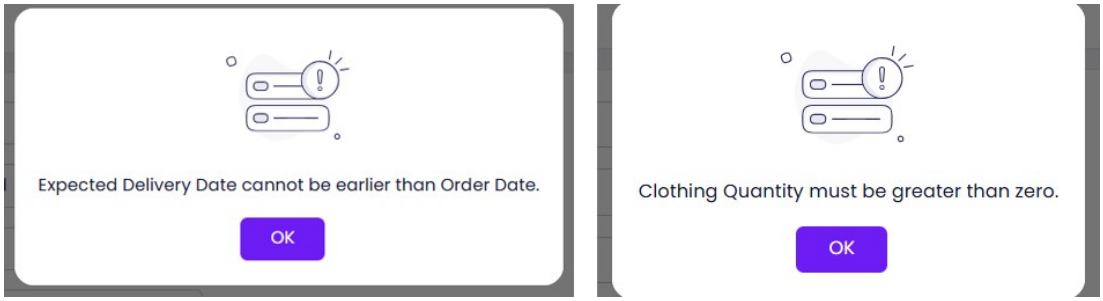


Figure 6.1: Validation Alert from the Validate Delivery Date Workflow

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	First notification email was not found in the inbox	Mail filter placed the Zoho Creator sender in Spam	Message located in Spam; later email received after editing an order	Closed
BUG-02	Dashboard chart titles did not match the data plotted	Titles generated with the first version did not match the configured axes	Titles corrected in the dashboard builder	Closed
BUG-03	Service Catalogue prices display a dollar symbol (224446) while orders display rupees	Currency setting of Price per Unit differs from the order form	Currency of Price per Unit in the Service Catalogue changed to INR	Closed

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-04	“Orders In Progress” card counts only Washing orders	Card criteria is Order Status = Washing	None applied. Suggested correction: add Ironing to the card criteria or rename the card	Open – Pending Verification
BUG-05	Customer appears as an ID (for example 1) in the order form, reports and email (224516, 224433, 224757)	Customer lookup displays Customer ID	None applied. Suggested correction: change the lookup display field to Customer Name	Open – Pending Verification
BUG-06	Unit prices of the sample orders differ from the Service Catalogue prices (for example, Order 1 shows ₹25.50 for UrbanWash Express Wash against 2.50 in the catalogue, screenshot 224446)	Sample orders were generated separately from the catalogue	None applied. Suggested correction: align the sample orders with the catalogue prices (the orders are demonstration data)	Open – Pending Verification

BUG-03 (currency symbol) was closed after the Service Catalogue currency was changed to INR; it is separate from BUG-06, which concerns the unit-price values. BUG-04, BUG-05 and BUG-06 remain open and pending verification; no evidence of a fix has been recorded.

6.5 User Acceptance Testing and Feedback

To obtain preliminary user feedback, the UrbanWash Laundry Service Order Management application was demonstrated to two individuals. Their observations focused on the application’s usability, usefulness and ability to provide information about laundry orders.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Raj	Viewer of a demonstration of the application	Found it a useful way to track laundry orders digitally; such a tracking facility is not readily available in his usual laundry experience; felt it could help monitor the progress of laundry services.	No changes made within the project timeline; feedback recorded for consideration in future enhancements (Section 9.2).

User	Role Tested	Feedback	Action Taken
Sam	Viewer of a demonstration of the application	Found the application simple and easy to use; appreciated convenient access to information about clothing and laundry orders; wanted clearer information about the status and handling of her clothes; also asked whether the application was safe.	No changes made within the project timeline; feedback recorded for consideration in future enhancements (Section 9.2).

Feedback from User 1 – Raj. Raj observed that the application provides a useful way to track laundry orders digitally. He mentioned that such a tracking facility is not readily available in his usual real-world laundry experience and felt that the application could be helpful in monitoring the progress of laundry services.

Feedback from User 2 – Sam. Sam found the application simple and easy to use. She appreciated the ability to access information about clothing and laundry orders conveniently. She also expressed interest in having clearer information about the status and handling of her clothes through the application, and asked whether the application was safe. As stated in Section 7.3, only platform-level controls (Zoho account login and HTTPS) were identified, permission enforcement was not verified and no security assessment was carried out.

Overall Feedback. The feedback indicated that the application was perceived as simple to navigate and potentially useful for improving the visibility of laundry order information. Both users recognised the value of having a centralised system for tracking orders instead of relying entirely on manual enquiries.

These observations represent preliminary qualitative feedback from two individuals and should not be interpreted as a formal usability study or a comprehensive security assessment. No ratings, percentages or survey scores were collected.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

UrbanWash was built and run in Zoho Creator's web-based development environment under a trial account (the edition was not verified), and it is reached through its Zoho Creator application URL. It was demonstrated in this trial environment. No separate Stage or Production environment, version notes or rollback plan is claimed, and the sample records were not removed, so the application is a development and demonstration deployment rather than a production release. Table 7.1 records the status of each checklist item.

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (√)
1	All test cases executed in Development / Stage	No – 9 of 16 executed
2	Sample/test data removed or isolated	No – sample data retained
3	Roles, permissions and users configured in Production	N/A – no Production environment
4	Connections authorised in Production	N/A – no connections used
5	Schedules and workflows enabled	Yes – five form workflows
6	Application published and URL verified	Partial – URL accessible in the trial environment only
7	Version / release notes recorded	No

7.2 Performance Optimisation

No performance measurements, benchmarks or load tests were carried out, and the application holds only a handful of sample records. The only technique worth noting is

that Auto Price retrieves the service with a single fetch by record ID instead of searching or looping; its effect was not measured.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Single fetch by record ID in Auto Price	Auto Price (DS-01)	Not measured	Not measured	Not measured

7.3 Security Features

Only the controls below were identified. Record-level permission criteria, encrypted fields, OAuth connections, IP or domain restrictions and two-factor authentication were not configured or verified in this project, so the application should not be described as fully secure.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Zoho account login	Access to the application URL	Access by unauthenticated users
CEO role and permission entries (Business Owner, Front Desk Staff, Read, Write)	Users → Roles and Permissions	Unauthorised data access (defined; assignment and enforcement not verified)
HTTPS application URL (platform feature)	Application URL	Interception of data in transit

7.4 Governance and Administration

The application was built and maintained by one developer who owns the Zoho Creator trial account. Forms, fields, reports and workflows use descriptive names (for example Auto Price and Validate Delivery Date), and the five workflows were created on 31 October 2026. Audit trail, change history, backup or export, user licence management and release notes were not implemented within the scope of the current project and are considered as possible future enhancements.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The following scenario follows one order through the application. The Front Desk Staff opens the Laundry Order form, selects a registered customer and a service, and the Unit Price appears automatically; after the quantity is entered the Total Amount is calculated, and the order is saved (Figure 8.1). When the laundry is finished, staff edit the order and set the Order Status to Ready for Delivery, and the alert email is sent (Figure 8.2). The Business Owner then opens the dashboard, where the KPI cards, charts and embedded report reflect the updated data (Figure 8.3). With the sample data, the dashboard showed ₹330 revenue from paid orders, 5 total orders and 2 orders ready for delivery, while the total value of all five orders was ₹1,476.

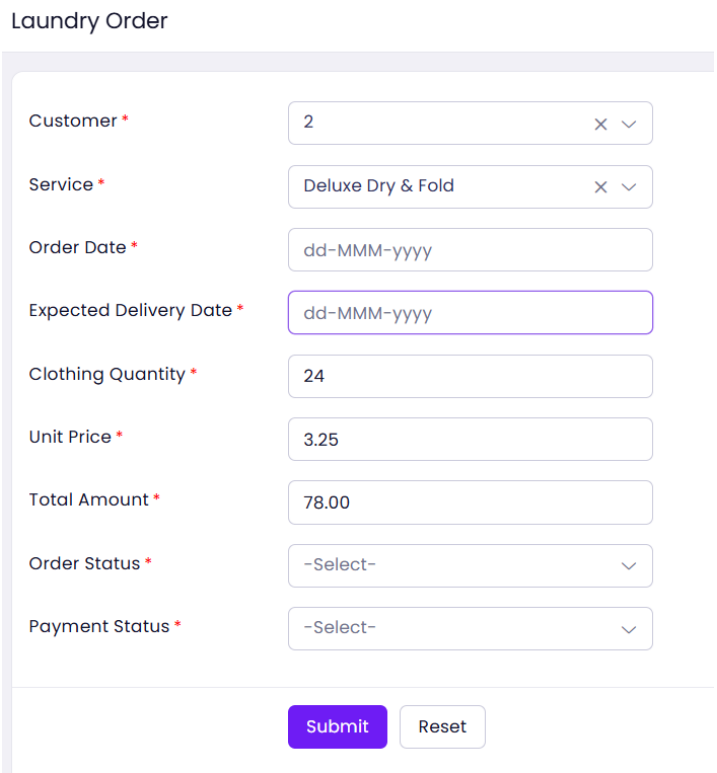


Figure 8.1: Scenario Step 1 – Creating an Order with Automatic Price and Total

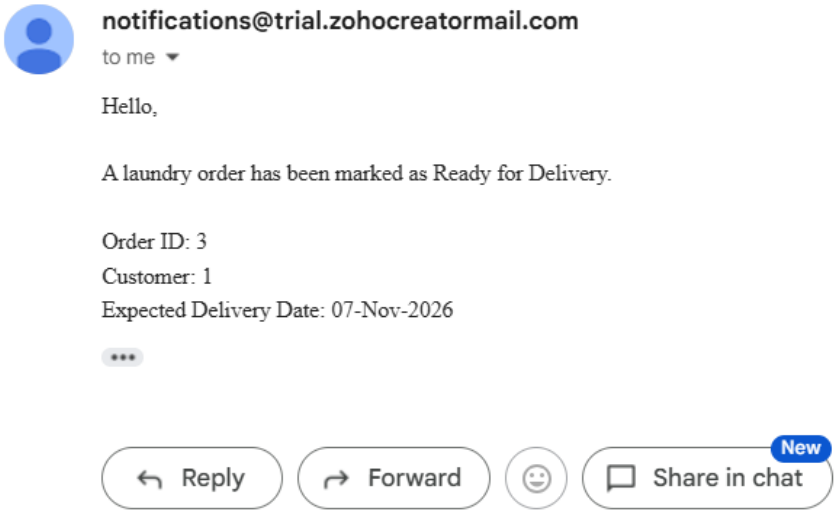


Figure 8.2: Scenario Step 2 – Ready for Delivery Email Alert

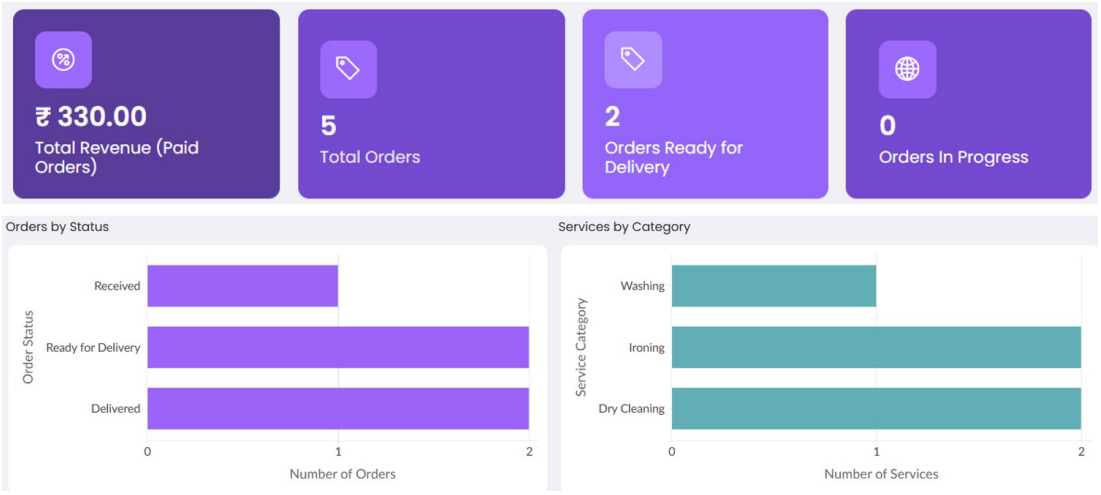


Figure 8.3: Scenario Step 3 – Dashboard After Order Updates

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Design a three-form data model with lookups	Sections 4.2, 5.2	Customer, Service Catalogue and Laundry Order forms; Customer and Service lookups work
Automate unit price and total using Deluge and workflows	Sections 5.5, 5.6	Unit Price and Total Amount are filled automatically (TC-04, TC-05)

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Validate delivery date and quantity	Sections 5.5, 5.6	Delivery-date and zero-quantity validation confirmed (TC-08, TC-10); negative-quantity case (TC-11) not executed
Notify by email when an order is Ready for Delivery	Section 5.6	Email received with order values (TC-13)
Provide reports and a dashboard	Sections 5.3, 5.4	Five report views; dashboard with 4 KPI cards and 2 charts showing live values (TC-14, TC-15)

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Pricing and total for one order	Price looked up and total calculated by hand	Unit Price and Total Amount filled automatically	Fewer manual calculations; time saved not measured
Manual steps	Customer and service details written for every order	Customer and service chosen from lookups	Less repeated entry and fewer spelling differences
Report generation	Totals and counts tallied by hand	Reports and dashboard computed from stored records	Revenue, order counts and status distribution available on demand
Data accessibility	Paper register at the counter	Records stored in Zoho Creator and opened in a browser	Records searchable from any permitted device
Customer notification	Staff call or message the customer	Email alert when an order is Ready for Delivery, sent to the Zoho login user and one additional configured address	Automatic alert to staff; Customer appears as an ID in the email

8.4 Limitations of the Low-Code Approach

Several limits became visible while building UrbanWash. The platform’s dashboard criteria are simple, so the Orders In Progress card counts only one status (Washing) and does not include Ironing; the workaround is to rename the card or extend its criteria. The Customer field inserted into the email shows the lookup ID instead of the customer’s

name, so the display value must be fetched explicitly if names are needed. Zia generated a usable first version, but its output contained mismatches (such as chart titles and currency symbols) that had to be found and corrected manually, so AI-generated structure still needs human verification. The application depends on the Zoho Creator platform for hosting, mail delivery and features, which is a form of vendor lock-in, and the notification email was sent from a Zoho trial sender that a mail filter treated as spam. Finally, the features that go beyond this project (integrations, payment, role-based access) would need more configuration or pro-code work. The three open defects are listed in Table 6.2 and their corrections are proposed in Section 9.2.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

On the no-code → low-code → pro-code spectrum, most of UrbanWash is no-code configuration: the three forms, their field types, the lookups, the five report views, the dashboard cards and charts, and the email action. Low-code is needed for the four Deluge scripts, which fetch the price, calculate the total and validate input. No pro-code was needed. A move to pro-code would be justified if the laundry needed an external payment gateway, a customer-facing mobile experience with a custom interface, or heavy data processing beyond what workflows handle comfortably.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

UrbanWash demonstrates that a small laundry's order process can be digitised on Zoho Creator with three linked forms, five report views, a dashboard and five form workflows. Staff select a customer and a service, the unit price and total are filled automatically, invalid delivery dates and quantities are rejected, orders are tracked through five stages, and an email alert is sent when an order is ready. The dashboard separates revenue from paid orders (₹330 in the sample data) from the total order value (₹1,476).

The application was built individually, starting from a Zia-generated structure that was then configured, corrected and tested manually. Nine of the sixteen test cases passed and seven were not executed within the project timeline; role-based permissions were not verified, and three defects remain open (Table 6.2). These limitations are stated openly in Chapters 6 and 8.

9.2 Future Enhancements

- Role-based permission profiles assigned to users and tested, so that the Business Owner and Front Desk Staff see only the components they need.
- Correction of the open defects: include Ironing in the Orders In Progress card, display Customer Name instead of Customer ID in the forms, reports and email alert, and align the sample order unit prices with the Service Catalogue.
- A way for customers to see the status of their orders (for example a customer portal or customer notifications), in response to the feedback in Section 6.5.
- Audit trail, backup and a separate Stage and Production deployment as governance enhancements (Chapter 7).
- Execution of the seven pending test cases and of role-based access tests.

9.3 Learning Outcomes and Career Relevance

Building UrbanWash gave practical experience of data modelling with lookups, form and dashboard design, Deluge scripting, workflow events (on user input, on validate and on successful submission), email automation and functional testing. It also showed

the importance of checking AI-generated output and of testing on the live application rather than only in the builder.

These skills relate to roles such as citizen developer, low-code developer and solutions consultant, where business processes are turned into working applications quickly, and they are a foundation for automation engineering and for learning pro-code extensions later.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Done	Application created with Zia and configured; sidebar sections	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Done	Customer, Service Catalogue and Laundry Order forms (12 distinct field types)	5.2	Fig. 5.2, 5.3; Table 4.2
II	Lookup field (relationship between forms)	Done	Laundry Order → Customer and Service Catalogue	4.2, 5.2	Table 4.1; Fig. 4.2
II	Field validations / layout rules	Done	Validate Delivery Date and Validate Quantity workflows	5.2	Fig. 6.1; Code 5.3
II	Kanban / Calendar / Timeline / Map report	Done	Order Statuses and Service Categories (Kanban)	5.3	Table 5.1
II	Page / dashboard built with UI Builder	Done	UrbanWash dashboard	5.4	Fig. 5.5
II	Deluge – form events (validate / success / field actions)	Done	Auto Price, Calculate Total (on user input); validations (on validate)	5.5	Code 5.1–5.3
II	Email / in-app notifications	Done	Ready for Delivery Alert (email)	5.5, 5.6	Fig. 5.6, 8.2
II	Form workflow	Done	Five form workflows	5.6	Table 5.3; Fig. 5.6
III	Zia / AI feature	Done	Create using Zia generated the first version of the application	5.9	Section 5.9
III	Analytics (charts / Zoho Analytics)	Done	Dashboard KPI cards and two charts	5.9	Fig. 5.8
V	Low-code limitations & spectrum analysis	Done	Sections 8.4 and 8.5	8.4, 8.5	Sections 8.4, 8.5

Total features implemented: 12

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7, 5.8	1	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	1	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	2	

Justification of the proposed levels (the examiner may revise them):

CO1 (Level 3). Chapters 1 and 2 explain low-code principles, compare them with traditional development and cite literature.

CO2 (Level 3). Three forms, five report views, a dashboard, four Deluge scripts and five form workflows were built and tested (Chapters 4 to 6).

CO3 (Level 1). No integrations or APIs were implemented; one role and four permission entries were defined but not verified.

CO4 (Level 1). The application ran only in the development environment of a trial account; no Stage or Production deployment, performance test or audit trail.

CO5 (Level 2). Zia-assisted building was applied and checked, and low-code limits were discussed (Sections 5.9, 8.4, 8.5); no pro-code extension.

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Laundry Order – Auto Price (DS-01): Created or Edited → User input of Service

```
if(input.Service != null)
{
    serviceRecord = Service_Catalogue[ID == input.Service];
    if(serviceRecord != null)
    {
        input.Unit_Price = serviceRecord.Price_per_Unit;
        if(input.Clothing_Quantity != null)
        {
            input.Total_Amount = input.Clothing_Quantity *
input.Unit_Price;
        }
    }
}
```

C.2 Laundry Order – Calculate Total (DS-02): Created or Edited → User input of Clothing Quantity

```
if(input.Clothing_Quantity != null && input.Unit_Price != null)
{
    input.Total_Amount = input.Clothing_Quantity * input.Unit_Price;
}
```

C.3 Laundry Order – Validate Delivery Date (DS-03): Created or Edited → Validations on form submission

```
if(input.Expected_Delivery_Date < input.Order_Date)
{
    alert "Expected Delivery Date cannot be earlier than Order Date.";
    cancel submit;
}
```

C.4 Laundry Order – Validate Quantity (DS-04): Created or Edited → Validations on form submission

```
if(input.Clothing_Quantity <= 0)
{
    alert "Clothing Quantity must be greater than zero.";
    cancel submit;
}
```

C.5 Laundry Order – Ready for Delivery Alert (Send Mail action): Edited → Successful form submission

```
Condition: Order_Status == "Ready for Delivery"
From: zoho.adminuserid
To: zoho.loginuserid and one additional configured email address
Subject: UrbanWash – Order Ready for Delivery
Message:
Hello,

A laundry order has been marked as Ready for Delivery.

Order ID: ${Order_ID}
Customer: ${Customer}
Expected Delivery Date: ${Expected_Delivery_Date}

Please check the UrbanWash application for further details.

Regards,
UrbanWash Laundry Services
```

Note: the scripts in C.1 to C.4 are transcribed from screenshots of the saved workflows (Auto Price, Calculate Total, Validate Delivery Date, Validate Quantity). DS-01 is the final combined version; it also recalculates Total Amount. C.5 shows the configuration of the Ready for Delivery Alert as it appears in the Send Mail action; it is a workflow action, not a Deluge script.

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the Zoho Creator login page in a web browser and sign in with your Zoho account. Open the UrbanWash Laundry Service Order Management application from your workspace. The sidebar shows Dashboard, Laundry Orders, Service Catalogues and Customers. Access through the Zoho Creator mobile app was not tested.

D.2 Instructions for the Front Desk Staff

- Open Customers → Customer, enter the customer's details and submit. Existing customers are listed in All Customers.
- Open Service Catalogues → Service Catalogue to add or review services and their price per unit.
- Open Laundry Orders → Laundry Order. Select the Customer and the Service; the Unit Price appears automatically.
- Enter the Order Date, Expected Delivery Date and Clothing Quantity; the Total Amount is calculated. The delivery date must not be earlier than the order date and the quantity must be greater than zero.
- Select the Order Status and Payment Status and submit the order.
- To update an order, open All Laundry Orders, edit the record, change Order Status (for example to Ready for Delivery) or Payment Status, and save. Setting Ready for Delivery sends the email alert.

D.3 Instructions for the Business Owner

- Open the Dashboard to see Total Revenue (Paid Orders), Total Orders, Orders Ready for Delivery and Orders In Progress.
- Read the Orders by Status and Services by Category charts for the distribution of orders and services.
- Use the embedded All Laundry Orders report or the Order Statuses report for details, and read the ready-for-delivery email for new alerts.